MILITARY SPECIFCATION

PELLET, FLASH TUBE, IB52

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements and quality assurance provisions and packaging for the IB52 Pellet for use in M788, M789, M799 and PXR6317 series of 30mm ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-3464	-	Desicants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-82484	-	Adhesive and Sealing Compounds, Cellulose Nitrate Base, Solvent Type

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-286	-	Propellants, Solid, Sampling, Examination and Testing
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGING DRAWINGS

11825947	-	Cartridge, 30mm, Target Practice, M788
11825919	-	Ignitor Compostion, IB-43
11825918	-	Fluid Ball Propellant, Type A
11825904	-	Pellet, Flash Tube, IB52

PUBLICATIONS

AMCR-715-505	Ammunition Balistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

CODE OF FEDERAL REGULATIONS

Title 49 Transportation

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of the conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

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3.2 IB52 pellet. The IB52 Pellet shall comply with all requirements of Drawing (Dwg) 11825904, all associated drawings, and with all requirements in applicable specifications.

3.3 Chemical properties. The IB52 Pellet material shall have the following chemical properties:

Property	Requirement
Moisture and volatiles	0.5 percent (maximum)
Calorific value	1200 calories per gram (min) 1550 calories per gram (max)

3.4 Particle size. The particle size shall be such that 3 percent maximum shall be retained by a US standard sieve No. 80 and 5 percent maximum shall pass through a US standard sieve No. 120.

3.5 Pellet crush strength. The average crush strength shall be no less than 22 Newtons.

3.6 Performance characteristics. The IB52 Pellet shall ignite the main propellant charge of a 30mm cartridge and shall provide the following ballistic performance of the target practice round (Dwg 1825947):

3.6.1 Ballistic performance requirements. The ballistic performance requirements of the IB52 Pellet when assembled and tested with the target practice round (11825947), as defined in 4.5.2.1 and 6.3.4, shall be as specified below.

TABLE I. Ballistic performance requirements

Characteristic	Temperature		
	-54°C	21°C	71°C
Case mouth pressure (mean peak)	1/	270-310 MPa	1/
Muzzle velocity (mean)	1/	795-815 M/S	1/
Standard deviation - maximum	15 M/S	15 M/S	15 M/S
Action time (maximum)	4.0 M/S	4.0 M/S	4.0 M/S
Mean + 3 standard deviation maximum	4.0 M/S	4.0 M/S	4.0 M/S

1/ Record for information only.

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3.6.2 Environmental conditions. When assembled into a complete round, as shown on Drawing 11825947 the pellet shall not deteriorate and shall meet the performance requirements defined in Table I, after being subjected to the following conditions:

3.6.2.1 Temperature, humidity and altitude. Temperature, humidity and altitude cycling conditioned with the temperature ranging from -54°C to 65°C, relative humidity up to 95 percent and pressure conditions equivalent to 15.2 kilometers altitude.

3.6.2.2 Aircraft vibration. Vibration as specified MIL-STD-810, method 514, except as modified in Section 4.

3.7 Processing. The material shall be processed in conformance with procedures which have been approved by the Government for the manufacture of this material. The contractor shall demonstrate that the process is controlled and repeatable. The Government shall be notified of any planned changes to the process and shall approve any changes prior to implementation. The contractor shall develop and maintain a complete record of the product process history of each lot or sublot to assure traceability. Such records shall be available for Government review.

3.7.1 Blending. The material components listed in 3.2 shall be blended in accordance with supplier generated procedures suitable for the manufacture of this material.

3.7.2 Uniformity. The material components shall be blended such that a uniform mixture is obtained. A uniform lot is one that does not exhibit layering of the orange layers of ball propellant through the dark IB52 mix or segregation of the constituents.

3.7.3 Pelletizing. The pellet configuration material shall be formed in accordance with supplier generated procedures suitable for the manufacture of this material and shall conform to Drawing 11825904.

3.8 First article inspection. This inspection contains technical provisions for first article inspections. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The material shall be manufactured in accordance with best commercial practice. Care shall be exercised to minimize contaimination during manufacture, packing and storage. Cracked pellets and foreign material are not permitted. Chipped pellets meeting weight and dimensional requirements are allowed.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection. (Table II)

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in quantities as indicated.

Part Description	Drawing	Quantity
Pellet, Flash Tube IB52	11825904	400
IB52 (Prior to Pelleting)		58 grams

4.3.1.1 Contractor shall use 300 of the above pellets to assemble 100 30mm M788 TP cartridges in accordance with Dwg 11825947. Component parts for the assembly will be provided as GFM. The remaining 100 pellets will be submitted as is.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table II.

4.3.2.1 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract drawings and specification requirements. The contractor's inspection shall be subjected to verification by the Government representative.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63986 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Pellet, IB52				11825904
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
A 1/ 3/	Uniformity (granular)	50 grams	ACC-0 REJ-1	3.7.2	4.4.3.2
A	Particle size (granular)		ACC-0 REJ-1	3.4	4.5.3.1
A	Chemical composition (granular)		ACC-0 REJ-1	3.2	4.5.1.1
	Moisture & volatiles (granular)	3 grams	ACC-0 REJ-1	3.3	4.5.2.1
	Crush strength (pellets)	30	ACC-0 REJ-1	3.5	4.5.3.2
	Dimensions (pellets)	30		3.2	
	Outside diameter		ACC-0 REJ-1		
	Length		ACC-0 REJ-1		
	Weight		ACC-0 REJ-1		

NOTES:
1/ Tests identified by the same letter may be conducted on the same test items.
3/ Test A should be done in the following order: uniformity, particle size and chemical composition.

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TABLE II. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63986 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Pellet, IB52				11825904
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
	Workmanship (pellet)	225 grams	ACC-0 REJ-1	3.9	4.4.3.3
B 1/	Case mouth pressure	20 2/		Table I	4.5.4.8
B	Muzzle velocity			Table I	4.5.4.9
B	Action time			Table I	4.5.4.10
	Temperature, humidity, altitude	20	ACC-0 REJ-1	3.6.2.1	4.5.4.11.1
	Vibration	20	ACC-0 REJ-1	3.6.2.2	4.5.4.11.2
	Calorific value	5grams	ACC-0 REJ-1	3.3	4.5.2.2

NOTES:

2/ Sample quantity at each temperature, -54°C, 21°C and 71°C

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4.4 Quality conformance inspection

4.4.1 Inspection for formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63986 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Pellet Material, IB52 (Granular prior to pelletizing)				11825904
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Chemical uniformity	50 grams 1/		3.7.2	4.4.3.2
102	Particle size			3.4	4.5.3.1
103	Chemical composition			3.2	4.5.1.1
104	Calorific value	5 grams		3.3	4.5.2.2
105	Moisture and volatiles	3 grams		3.3	4.5.2.1
Minor	None defined				

NOTES: 1/ The sample shall be used for chemical uniformity, particle size, and chemical composition

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63986 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825904
4.4.2.2	Pellet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight		0.25%	3.2	Balance
102	Crush strength		0.25%	3.5	4.5.3.2
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Length		0.65%	3.2	Gage
203	Workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63986 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Pellet (after assembly into cartridge)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
Critical	None defined				
Major					
101	Case mouth pressure (hot, cold, ambient)	20 1/ 2/		Table I	4.5.4.8
102	Muzzle velocity (hot, cold, ambient)			Table I	4.5.4.9
103	Action time (hot, cold, ambient)			Table I	4.5.4.10

NOTES:
1/ Sample size applies to each temperature, - 54°C, 21°C and 71°C
2/ Sample shall be used for all three tests (obtain data simultaneously)

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4.4.3 Testing.

4.4.3.1 Material tests. Two containers shall be selected at random from the inspection lot. Approximately 0.5 kg samples shall be removed from each selected container. The sample shall be subjected to the chemical composition, chemical properties and physical characteristics tests. If any sample fails to comply with the requirements, the lot shall be rejected.

4.4.3.2 Uniformity. The IB52 Ignitor Mix shall be visually examined prior to pelletizing to note that the mix fluid ball powder and IB-43 material are uniformly mixed. The sample shall comply with the requirements of 3.7.2.

4.4.3.3 Workmanship. A 225 gram sample of the material, selected at random, shall be visually examined for evidence of broken or cracked pellets as well as foreign matter. Granular material shall be examined for presence of foreign material. The sample shall comply with the requirements of Table I. The total weight of fractured pellet pieces greater than 0.013 gram and less than 1.0 gram shall not be greater than 1.08 grams.

4.4.4 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and paragraph 6.4 herein.

4.5 Test methods and procedures.

4.5.1 Chemical composition. of IB52 material.

4.5.1.1 Composition weight. The percentage by weight of fluid ball propellant, Type A and 1B-43 ignitor mix for each lot shall be determined. The percentage of each shall be calculated to the nearest one-tenth of one percent by the following equations as applicable.

$$\text{Percent fluid ball propellant, Type A} = \frac{A \times 100}{A + B}$$

$$\text{Percent IB-43} = \frac{B \times 100}{A + B}$$

where

A = weight of fluid ball propellant, Type A in grams

B = weight of IB-43 in grams

These values shall comply with the requirements of 3.3.1.

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4.5.1.2 Certification.

4.5.1.2.1 IB-43. Certification of compliance to Drawing 11825919, Ignitor Composition, IB-43, shall be provided.

4.5.1.2.2 Fluid ball propellant, Type A. Certification of compliance to Drawing 11825918, Fluid Ball Propellant, Type A, shall be provided.

4.5.2 Chemical properties.

4.5.2.1 Moisture and volatiles. Add 2 to 3 grams of the 1B52 samples to a previously dried and tared weighing bottle, 30mm x 50mm, low form, Fisher Scientific No. 3-420, 35 ml capacity or disposable equivalent. Stopper the bottle and weigh the bottle and its contents to the nearest milligram. Remove the stopper and place the unstoppered bottle and sample in a nonforced draft oven controlled at a temperature of 100°C to 104°C for two hours. Remove from oven, reseat the stopper, cool the bottle and contents in a desiccator for 30 minutes to room temperature and weigh to the nearest milligram. Calculate the loss in weight as percent moisture and volatiles by the following equation.

$$\text{Percent moisture and volatiles} = \frac{(A - B) \times 100}{C}$$

where

A = weight of stoppered bottle and sample before heating, gm

B = weight of stoppered bottle and sample after heating, gm

C = Original weight of sample, gm

The percent moisture and volatiles shall conform to the requirements of 3.2.2.

4.5.2.2 Calorific value test. A sample of the material, prior to pelletization, shall be selected for use in the tests defined below.

4.5.2.2.1 Apparatus. The following apparatus shall be used for determining calorific value:

Adiabatic Calorimeter - Parr Model 1221 or equivalent

Double Value Oxygen Bomb - Parr Model 1101 or equivalent

Fuse Wire - Parr Model 45C10 or equivalent

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Thermometers (two) - 10-35°C with 0.03°C subdivisions, mercury type, calibrated every 1.4°C

Nitrogen - Dry nitrogen with regulator, gage and connector for pressurizing Parr bomb

4.5.2.2.2 Procedure. Turn on by-pass valves for hot and cold water. Put 1 cc of water in the bomb. Cut 10 cm length of fuse wire (provided by Parr Instrument Co.) and fasten onto bomb holder by threading through provided holes and stretching and bending downward into a loop.

Weigh one to two gram sample of 1B52 to within 0.0001 g and put into bomb cup placed in holder in bomb. Place fuse wire so it does not touch cup but touches the top of the ignitor sample. Assemble bomb, put on top, place ring on top, screw on collar tightly by hand and close bomb valve with wrench supplied. Attach bomb with Parr gauge and connetions to nitrogen cylinder and fill bomb to 2.53 MPa with nitrogen the second time. Nitrogen must be pure, no oxygen, as heat of combustion in most cases is greater than heat or explosion. Weigh bucket and weigh into it 2000 g of distilled water. Place bomb in calorimeter with fingers and shake water back intobucket from fingers. Attach cable for foring fuse to the bomb. Before attaching cable check connection by contacting the cable to bomb lid and pressing fuse button. If pilot light burns, connection is good. Close calorimeter, start stirrers and lower thermometers into water. Adjust jacket temperature to the same as the bucket temperature by running hot or cold water into the jacket. When the bucket temperature is constant, three minute interval readings the same, fire ignitor composition by pressing fuse button for 4 seconds or until pilot light goes out. Take bucket temperature every minute and adjust jacket temperature to bucket temperature. Jacket temperature can be 0.1°C from bucket temperature as bucket temperature rises but must be the same as start and end of temperature rise. When three minute interval bucket temperatures are the same the final temperature is reached. Open calorimeter, by shutting off stirrer, raising thermometer and sliding lid around. Remove bomb by taking off fuse cable and lifting out of bucket. Release pressure by opening bomb valve and open bomb by unscrewing collar and lifting off lid. Remove unburned part of fuse and measure.

4.5.2.2.3 Calculations.

$$\text{Calorific value, cal/g} = \frac{(T_f - T_o)F}{W}$$

where

W = weight of sample

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L = length offuse wire burned = 10-length of wire not burned

T_o = initial temperature reading (corrected, ^oC)

T_f = final temperature reading (corrected, ^oC)

F = bomb factor = 2484.5

Determine temperature correction for thermometer from chart. The calorific value shall conform to the requirements of 3.3.

4.5.3 Physical characteristics.

4.5.3.1 Particle size. The IB52 material shall be subjected to the granulation test of MIL-STD-286, Method 506.1. The calculated percentage value shall comply with the requirements defined in 3.4.

4.5.3.2 Pellet Crush strength . The pellet crush strength shall be evaluated using a Hunter Tester Model L30 or equivalent device. The pellet holding fixture shall be a plate with a 3.175 mm radius positioning slot. Alternate procedures must be approved by the Government. A single IB52 ignitor pellet shall be placed horizontally in the positioning slot on the pellet holding fixture. With the pellet in the fixture slowly apply a force to the pellet until break-up occurs. The maximum applied force shall be recorded and average value calculated.

4.5.4 Ballistic tests.

4.5.4.1 Test item manufacture. Unless otherwise specified, all tests shall be conducted on material which has been manufactured using the same materials, facilities, processes and procedures which are used in regular production. Ballistic and environmental tests shall be conducted on complete rounds which have been loaded and assembled in accordance with the same production and inspection processes as will be used in fulfilling production contract requirements and loaded at a loading facility approved by the procuring activity. Unless otherwise specified, the test item shall conform with the requirements of Drawing 11825947. Components for the assembly of the test items shall be from accepted lots.

4.5.4.2 Test item preconditioning. Test items to be subjected to temperature tests shall be assembled from components which have been conditioned at $21^o \pm 10^oC$ for 12 hours or the complete cartridge shall be subjected to the same pre-test conditioning.

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4.5.4.3 Test item conditioning. For test methods which require test item conditioning at -54°C, 21°C or +71°C, the item shall be exposed to the specific temperature for a period of not less than 12 hours. The conditioning time of +71°C shall not exceed 24 hours. The conditioning temperature shall be controlled to ±5°C.

4.5.4.4 Reference rounds. Reference rounds shall be fired in conjunction with ballistic tests. Unless otherwise specified, the reference rounds shall be assembled in accordance with the requirements of Drawing 11825947 and inspected in accordance with this specification with the charge weight controlled to ±0.2 gram. A minimum of 10 reference rounds shall be fired during each ballistic firing test and the mean muzzle velocity, case mouth pressure and action time recorded. Any individual round readings which vary from the mean by greater than 12 meters per second, 20 MPa or 0.3 millisecond respectively, shall be disregarded and a new mean value shall be computed. If more than three rounds are disregarded, additional reference rounds shall be fired, provided they are fired during the same day as the applicable tests are conducted. The differences between the mean reference round test values and the reference acceptance baseline values shall not exceed 10 meters per second, 23 MPa, or 0.35 millisecond.

4.5.4.5 Test gun. Functional tests shall be conducted on a single shot test barrel as shown on Drawing 11825969 or equivalent. The barrel used for ballistic testing shall not be used for testing if any of the following conditions exist.

a. The barrel does not meet print requirements.

b. Variation of reference lot muzzle velocity by more than 15 meters per second.

4.5.4.6 Test measurement equipment. Tests specified herein shall be conducted using test equipment having a precision equal to or better than that shown below.

a.	Muzzle velocity	±3 meters per second
b.	Case mouth pressure	±5.0 percent
c.	Action time	±0.10 ms

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4.5.4.7 Calibration of instrumentation. A reference lot of 30mm TP ammunition shall be manufactured. It will be tested and established as a control baseline. The reference lot accuracy will be established by averaging the first three 20 round accuracy tests. Prior to acceptance teting of each lot, a minimum of ten rounds per day will be fired to establish pressure, velocity and action time. Ballistic variation between refernce and acceptance tests lots will be adjusted as follows:

$$X_a + (X_{ra} - X_r) = \text{corrected } X_a$$

where

X_a is mean of accptance lot

X_r is mean of reference lot

X_{ra} is mean of acceptance baseline

4.5.4.8 Case mouth pressure. The test rounds shall be conditioned at -54°C, 21°C and 71°C temperatures and fired from a test fixture barrel. Pressure measurements shall be made at the case mouth with the piezoelectric transducer. A permanent record shall be made of the resulting pressure. This record may be a pressure trace, paper print out of the data or other similar instrument reading. The pressure shall conform to the requirements of Table I.

4.5.4.9 Muzzle velocity. Muzzle velocity shall be measured by the use of screens located a known distance apart (30 meters preferred). The test sample rounds shall be conditioned at -54°C, 21°C, and 71°C before firing. The mean velocity and the standard deviation from the mean of the sample rounds shall be calculated and shall comply with the requirement of Table I. The mean velocity data at -54°C and 71°C shall be recorded for information only; however, the standard deviation requirement shall be met. Velocity data shall be corrected to the muzzle condition using a positive correction value of +0.75 m/sec per meter multiplied by the distance between the muzzle and the midpoint of the velocity screens. See 6.3.1 for example of this procedure.

4.5.4.10 Action time. The test rounds shall be conditioned at -54°C, 21°C and 71°C and fired from a test fixture barrel. The time between application of voltage to primer and exit of the projectile from the muzzle shall be measured. The time shall comply with the requirement of Table I. Any corrected action time of 2.40 milliseconds or less shall be regarded as a potential instrumentation error and referred to the Government for disposition.

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4.5.4.11 Environmental tests. Unless otherwise pecified, performance during and/or after environmental exposure, as specified in the applicable test procedure, shall consist of conducting the tests of 4.5.4.8, 4.5.4.9 and 4.5.4.10. except all test rounds shall be conditioned at 21°C prior to firing.

4.5.4.11.1 Temperature, humidity and altitude. The test rounds shall be subjected to the temperature/humidity/altitude test of MIL-STD-810, Method 504. After exposure, the test item shall be examined for deterioration and subjected to the test referenced in 4.6.2.

4.5.4.11.2 Aircraft vibration. The test rounds shall be subjected to a broad band random vibration profile in accordance with the requirements of MIL-STD-810, Method 514.2, Procedure I, Category C, except the vibration spectrum shall be as shown in Figure 1. The test item shall be mounted to a test fixture simulating an AAH magazine in a normal airplane mounting as specified by the procuring agency. The test duration shall be 45 minutes in each of three mutually perpendicular axes at each of two temperature conditions, -40° ±2.8°C and 65.5° ±2.8°C. Following exposure, all of the test items shall be subjected to the tests referenced in 4.3.2.

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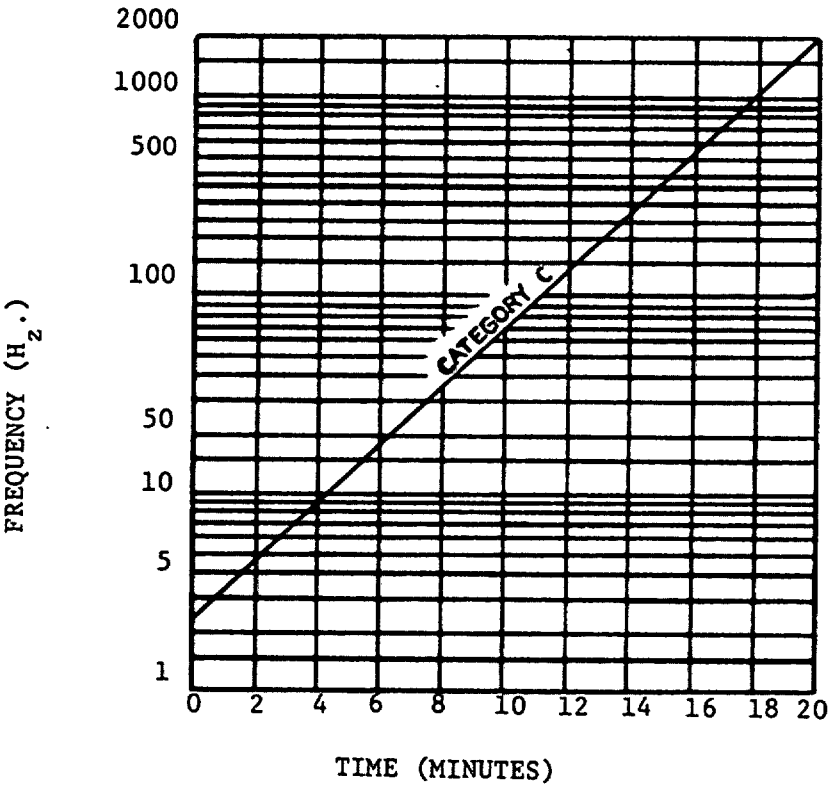


FIGURE 1. VIBRATION TEST REQUIREMENTS

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4.5.4.12 Test data. Test data shall be recorded in uncorrected form as well as after correction for reference round data in accordance with the calibration procedure of 4.5.4.7. The test data shall also include any anomalies which may occur during all ballistics tests. The standard deviation shall be calculated in accordance with 6.3.2. If any sample cartridge firing defect is found to have been caused by a defective test weapon, the firing defects shall not be classed as a failure. In such case, the defective test weapons shall be corrected or replaced and the test repeated in whole or in part, as appropriate. If such firing defect is not found to have been caused by a defective weapon, it shall be classed as a failure.

5.0 PACKAGING

5.1 Preservation, packaging and packing. The IB52 pellets shall be packaged in sealed, conductive plastic, containers for interplant transportation and storage. For shipment by common carrier the material shall be sealed in conductive plastic packages and further packaged in conformance with the requirements of Title 49, Code of Federal Regulations, Parts 100 through 199.

6.0 NOTES

6.1 Intended use. The IB52 pellet is intended for use as an ignition booster for the 30mm family of cartridges including M788, and M789 ammunition.

6.2 Ordering data. See MIL-A-48078.

6.3 Ballistic test notes.

6.3.1 Muzzle velocity correction example. A velocity of 800.2 meters per second is recorded in a test instrumentation configuration with the midpoint between the velocity coils located 30 meters from the muzzle. The muzzle velocity is calculated as follows:

Muzzle velocity = 800.2 + 0.75(30) = 823 meters per second

6.3.2 Standard deviation. The standard deviation for use in testing for projectile extraction, muzzle velocity, chamber pressure and action time shall be calculated in accordance with the following formula:

$$\sigma = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n - 1}}$$

6.4 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment design as required to: Commander, ATTN: DRDAR-QAF-I, ARRADCOM, Dover, NJ 07801

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6.5 Submission of results of contractor-conducted examination and tests. Unless otherwise specified by the contracting officer, the contractor should forward requested records of examination to: Commander, ATTN: DRDAR-QAF-S, ARRADCOM, Dover, NJ 07801

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DOD-P-63986 PELLET, FLASH TUBE, IB52

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